Coding Assignment for Category C

Problem 1: The Canteen Billing Counter

Task

A college canteen serves three kinds of customers: students, staff, and guests. Each kind of customer pays a different final amount for the same bill. For every bill, calculate the final amount to be paid, display it, and then display the total amount collected.

Input

The first line contains an integer N, the number of bills. Each of the next N lines contains a customer type and the bill amount.

- STUDENT Amount
- STAFF Amount
- GUEST Amount

Output

For each bill, display CustomerType: FinalAmount, formatted to two decimal places. Finally, display Total: GrandTotal, also formatted to two decimal places.

Business Rules

- Students get a 10% discount.
- Staff get a 5% discount.
- Guests pay the full amount plus a ₹10 service charge.

Constraints

- 1 <= N <= 1000
- 1 <= Amount <= 100000

Sample Input

```
3
STUDENT 200
STAFF 300
GUEST 150
```

Expected Output

```
STUDENT: 180.00
STAFF: 285.00
GUEST: 160.00
Total: 625.00
```

Instructions

Consider how to model different customer types that share a common operation (calculating the final amount) but follow different pricing rules. Focus on how the billing counter can process a list of bills from different customer types without checking the type of each customer with if-else blocks.

Problem 2: The Campus Parking Charge Calculator

Task

The campus parking lot allows bikes, cars, and trucks. Each vehicle type is charged differently based on the number of hours it was parked. For every parked vehicle, calculate the parking charge, display it, and then display the total charge collected.

Input

The first line contains an integer N, the number of parked vehicles. Each of the next N lines contains a vehicle type and the number of hours parked.

- BIKE Hours
- CAR Hours
- TRUCK Hours

Output

For each vehicle, display VehicleType: Charge, formatted to two decimal places. Finally, display Total: GrandTotal, also formatted to two decimal places.

Business Rules

- Bike: ₹10 per hour.
- Car: ₹30 for the first hour, plus ₹20 for each additional hour.
- Truck: ₹50 per hour, with a minimum charge of ₹100.

Constraints

- 1 <= N <= 500
- Hours is a whole number, 1 <= Hours <= 24

Sample Input

```
4
BIKE 3
CAR 4
TRUCK 1
CAR 1
```

Expected Output

```
BIKE: 30.00
CAR: 90.00
TRUCK: 100.00
CAR: 30.00
Total: 250.00
```

Instructions

Consider how to model different vehicle types that share a common operation (calculating the parking charge) but have their own charging rules. Focus on how the parking system can calculate charges for a list of different vehicles without explicit type checking.

Problem 3: The Hostel Electricity Bill

Task

A hostel has three kinds of rooms: single rooms, shared rooms, and AC rooms. Each room type calculates its monthly electricity bill differently. For shared rooms, the bill is split equally among the occupants, and the amount shown is what each occupant pays. For every room, calculate the bill, display it, and then display the total of all bills shown.

Input

The first line contains an integer N, the number of rooms. Each of the next N lines contains RoomType Units [AdditionalParam].

- SINGLE Units
- SHARED Units Occupants

- AC Units

Output

For each room, display RoomType: BillAmount, formatted to two decimal places. Finally, display Total: GrandTotal, also formatted to two decimal places.

Business Rules

- Single room: ₹8 per unit.
- Shared room: ₹6 per unit, divided equally by the number of occupants.
- AC room: ₹10 per unit, plus a fixed charge of ₹200.

Constraints

- 1 <= N <= 200
- Units is a positive whole number
- 2 <= Occupants <= 4

Sample Input

```
3
SINGLE 120
SHARED 150 3
AC 100
```

Expected Output

```
SINGLE: 960.00
SHARED: 300.00
AC: 1200.00
Total: 2460.00
```

Instructions

Consider how to model different room types that share a common operation (calculating the bill) but use different rules. Notice that a shared room needs one extra value (occupants); think about where that value should be stored so the main processing flow can still treat every room the same way.

Problem 4: The Festival Bonus Calculator

Task

A company pays a festival bonus to its employees. Full-time employees, part-time employees, and interns each receive the bonus in a different way. For every employee, calculate the bonus, display it with the employee's name, and then display the total bonus paid.

Input

The first line contains an integer N, the number of employees. Each of the next N lines contains EmployeeType Name MonthlySalary.

- FULLTIME Name MonthlySalary
- PARTTIME Name MonthlySalary
- INTERN Name MonthlySalary

Output

For each employee, display Name: Bonus, formatted to two decimal places. Finally, display Total Bonus: GrandTotal, also formatted to two decimal places.

Business Rules

- Full-time employees get 10% of their monthly salary.

- Part-time employees get 5% of their monthly salary.
- Interns get a fixed bonus of ₹2,000, whatever their salary.

Constraints

- 1 <= N <= 1000
- Name is a single word
- 1000 <= MonthlySalary <= 500000

Sample Input

```
3
FULLTIME Asha 50000
PARTTIME Ravi 30000
INTERN Neha 15000
```

Expected Output

```
Asha: 5000.00
Ravi: 1500.00
Neha: 2000.00
Total Bonus: 8500.00
```

Instructions

Consider how to model different employee types that share common details (name and salary) and a common operation (calculating the bonus), while each type follows its own bonus rule. Focus on how the payroll system can process a list of employees without explicit type checking.

Problem 5: The Streaming Plan Renewal Reminder

Task

A video streaming service offers three subscription plans: Basic, Standard, and Premium. Each plan stays valid for a different number of days from its start date. For every subscriber, calculate the date on which the plan must be renewed and display it.

Input

The first line contains an integer N, the number of subscribers. Each of the next N lines contains PlanType Name StartDate, where StartDate is in the format YYYY-MM-DD.

- BASIC Name StartDate
- STANDARD Name StartDate
- PREMIUM Name StartDate

Output

For each subscriber, display Name: RenewalDate, where RenewalDate is formatted as YYYY-MM-DD.

Business Rules

- Basic plan: valid for 30 days.
- Standard plan: valid for 90 days.
- Premium plan: valid for 365 days.
- The renewal date is the start date plus the plan's validity in days.

Constraints

- 1 <= N <= 100
- Name is a single word
- StartDate is a valid date

Sample Input

```
4
BASIC Asha 2024-01-15
STANDARD Ravi 2024-02-01
PREMIUM Neha 2024-03-10
BASIC Kiran 2024-12-20
```

Expected Output

```
Asha: 2024-02-14
Ravi: 2024-05-01
Neha: 2025-03-10
Kiran: 2025-01-19
```

Instructions

Consider how to model different plans that share a common operation (finding the renewal date) but have different validity periods. Focus on how the system can process a list of subscribers without explicit type checking. You may use java.time.LocalDate and its plusDays() method; note that the sample includes a leap year (2024) and a change of year.